

DEEP LEARNING: ADVANCED MODELS AND METHODS

# Knowledge Distillation for Mobile Action Recognition

Compressing a 3D ResNet-50 into an ultra-lightweight MobileNet3D on UCF-101 via logit-based, attention transfer, and cross-frame distillation.

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**TRACK**

Track 6 - KD for Mobile Action Recognition

**INSTITUTION**

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**REPOSITORY**

[github.com/danii909/ResNet-to-MobileNet-Action](https://github.com/danii909/ResNet-to-MobileNet-Action)

# The Capacity Gap

**Goal:** Compress a 3D ResNet-50 teacher (pretrained on Kinetics-400) into a MobileNet3D student for on-device video action recognition on UCF-101 (101 classes).

TEACHER

**3D ResNet-50**

**88.82%**

TEST ACCURACY

~128 MB

6.21 ms GPU

76.6 ms CPU

ACCURACY GAP

**-29.3 pp**

SIZE REDUCTION

**13.5×**

INFERENCE SPEED

**1.82×**

1.66× CPU (BS=1)

STUDENT

**MobileNet3D**

**59.48%**

TEST ACCURACY

~9.5 MB

3.42 ms GPU

46.1 ms CPU



# UCF-101 & Group-Aware Split

**101**  
ACTION CLASSES

**13,320**  
VIDEO CLIPS

**24f**  
@ 112×112 PX

## Original Split Structure

UCF-101 is officially divided only into Train and Test partitions, **lacking an actual validation set** required for proper model selection.

## ⚠ Data Leakage Risk

Correlated clips from the same source video share actor & background. Random splits cause **spatial leakage** and inflated accuracy.

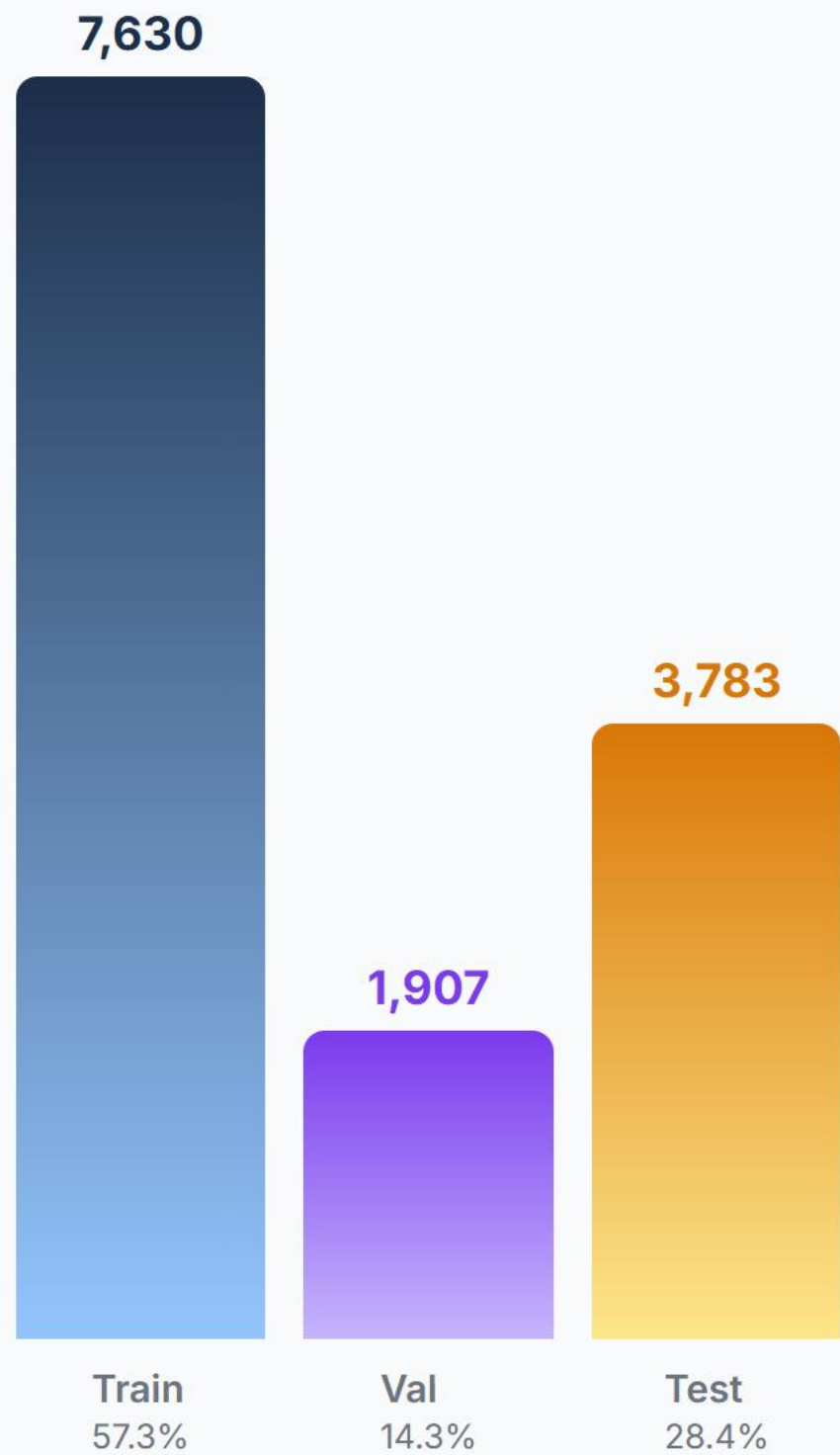
## ✅ Group-Aware Split

Isolation by group ID (**\_gXX\_**). No source video shared across Train / Val / Test.

## 🕒 Temporal Sampling & Stride

Segmented sampling divides the video into  $N$  segments, randomly sampling one frame per segment in training (uniform for eval). Stride is dynamic ( $\approx L / N$  frames) to span the entire clip.

## NEW SPLIT DISTRIBUTION



# Knowledge Distillation Loss

Balanced objective function for knowledge transfer

$$\mathcal{L}_{\text{total}} = \underbrace{\alpha \cdot T^2 \cdot D_{\text{KL}} \left( \sigma(z^S / T) \parallel \sigma(z^T / T) \right)}_{\mathcal{L}_{\text{soft}}} + \underbrace{(1 - \alpha) \cdot \mathcal{L}_{\text{CE}}(z^S, y)}_{\mathcal{L}_{\text{hard}}}$$

## $\alpha = 0.7$

### TARGET BALANCING

Always set to **0.7** for the majority of experiments to prioritize mimicking the Teacher's soft targets (70%) over the hard ground truth labels (30%).

## $T \in \{1, 5, 8, 10, 20\}$

### LOGIT SCALING

The main hyperparameter varied across experiments to control the smoothness of probability distributions, extracting latent inter-class similarities.



# Augmentation Trade-off & Data Leakage

All tests: T=10, a=0.7, 24f · Why light augmentation was chosen

## 1. Minimal Augmentation

WITH LEAK

**65.7%**

Top-5: 87.8%

NO LEAK

**59.3%**

Top-5: 83.6%

### CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Disabled (0.0)
- **Random Erasing:** Disabled
- **Temporal stride:** Fixed (max\_stride=1)

## 2. Light Augmentation ✓

WITH LEAK

**69.5%**

Top-5: 90.3%

NO LEAK

**64.5%**

Top-5: 87.4%

### CONFIGURATION

- **Spatial crop:** Random Crop (112×112)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Light (0.1)
- **Random Erasing:** 5% probability
- **Temporal stride:** Fixed (max\_stride=1)

## 3. Strong Augmentation

WITH LEAK

**55.83%**

Top-5: 83.27%

NO LEAK

**50.4%**

Top-5: 78.5%

### CONFIGURATION

- **Spatial crop:** Resized Crop (0.6—1.0)
- **Flips:** Horizontal Flip (p=0.5)
- **Color jitter:** Strong (0.2)
- **Random Erasing:** 15% probability
- **Temporal stride:** Variable (max\_stride=2)

# Temperature Sweep & Main Result

Systematic analysis of Student behavior as T varies · a = 0.7 fixed

| TEMPERATURE                  | BEST VAL ACC | EPOCH | TRAIN ACC | TEST TOP-1 | TEST TOP-5 | Δ VS BASELINE |
|------------------------------|--------------|-------|-----------|------------|------------|---------------|
| Minimal Augmentation (no KD) | 59.18%       | —     | 99.81%    | 59.48%     | 82.77%     | ref.          |
| T = 1                        | 62.75%       | 60    | 98.44%    | 62.91%     | 85.59%     | +3.43%        |
| T = 5                        | 63.12%       | 57    | 95.87%    | 62.07%     | 87.02%     | +2.59%        |
| T = 8                        | 64.57%       | 56    | 97.59%    | 64.31%     | 87.68%     | +4.83%        |
| T = 10                       | 64.85%       | 60    | 98.28%    | 64.47%     | 87.44%     | +4.99%        |
| T = 20                       | 65.18%       | 59    | 99.05%    | 65.85%     | 87.81%     | +6.37%        |

TEST ACCURACY VS. TEMPERATURE TREND





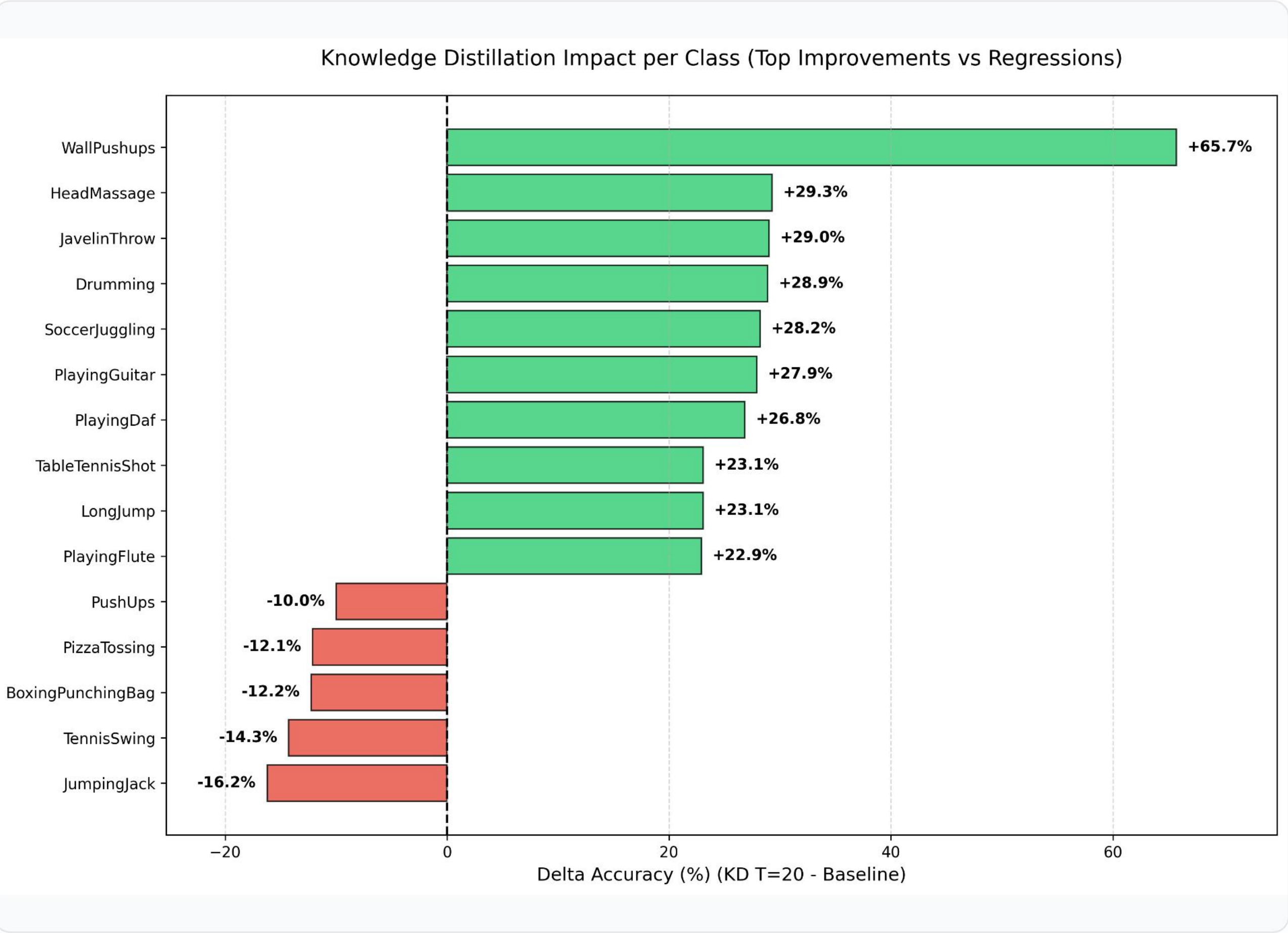
# Latent Space Analysis via t-SNE

2D projections of pre-classifier embeddings — 10 selected classes



The distilled model (T=20) shows significantly tighter and more distinct class clusters compared to the baseline student, successfully inheriting the well-separated semantic boundaries of the teacher model.

# Per-Class Accuracy: KD T=20 vs Baseline



### Improvements

Spatially complex classes (e.g. **WallPushups**, **HeadMassage**) gain most from the Teacher's visual filters.

### Regressions

Fast, cyclic actions (**JumpingJack**, **TennisSwing**) suffer from temporal over-smoothing at high T.

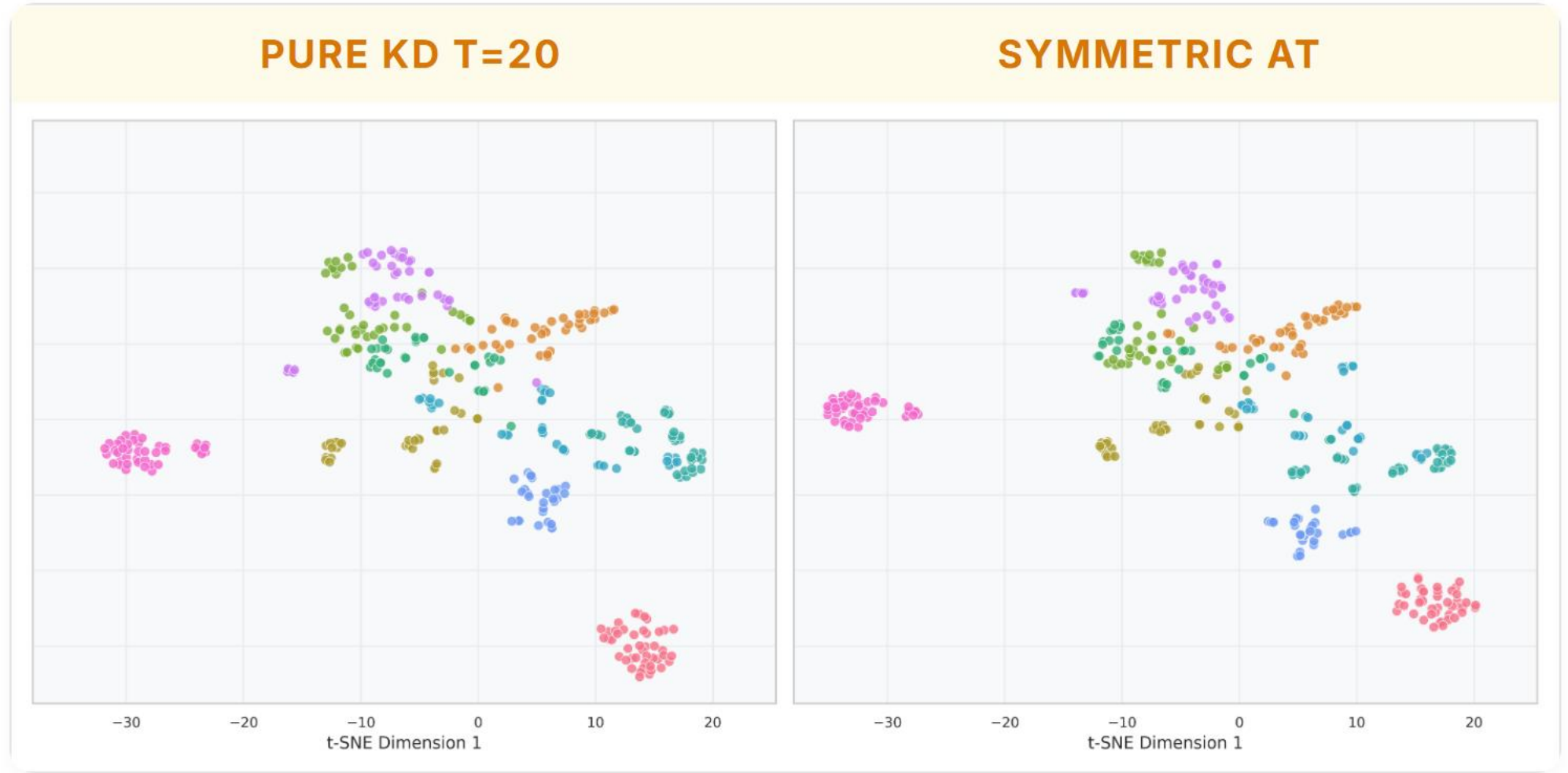
### Shared Challenges

Classes with high structural ambiguity (e.g. **HandstandWalking**, **Nunchucks**) or strong mutual confusion (e.g. **MoppingFloor**) remain unresolved in both models.

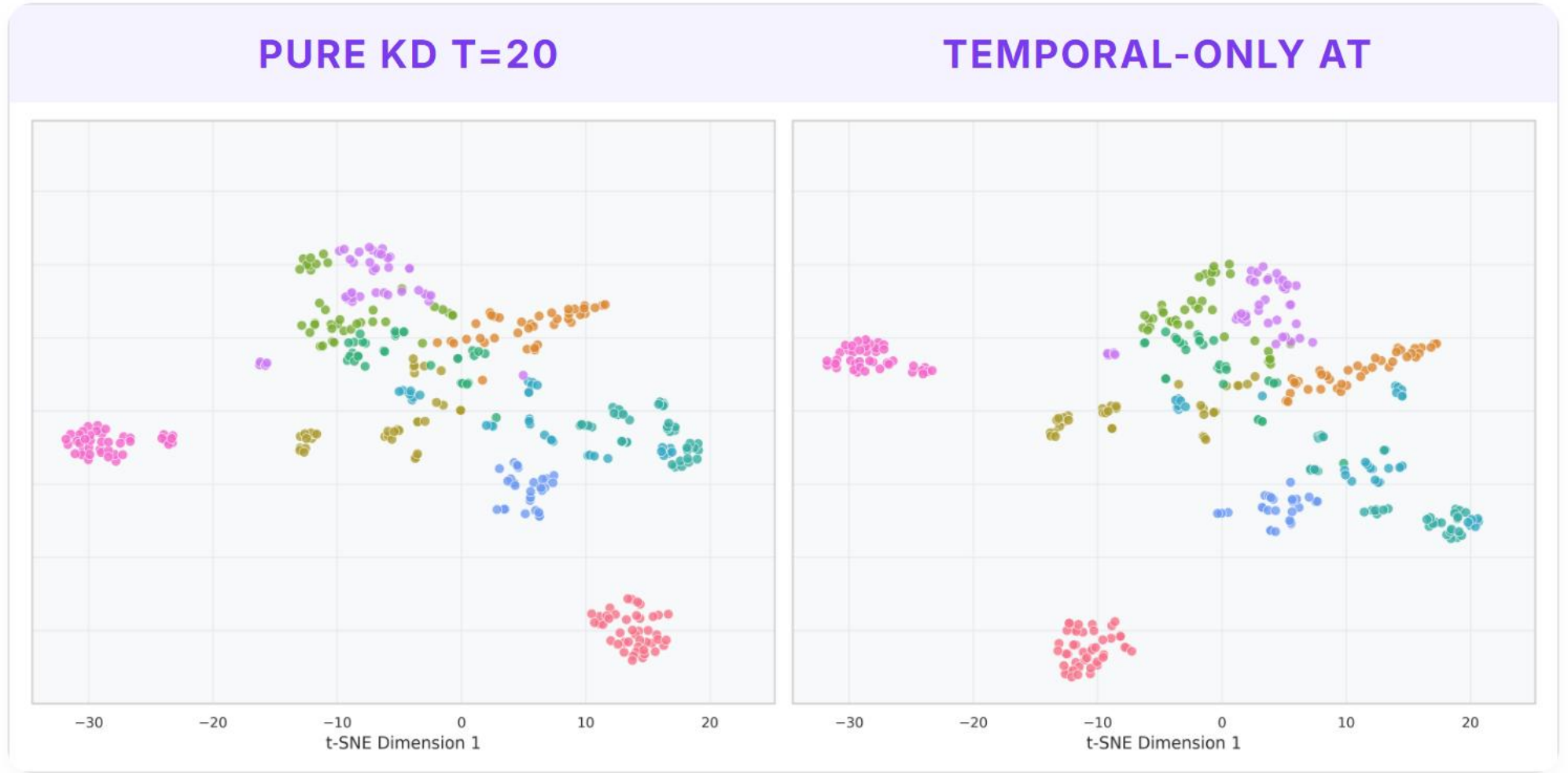


# Attention Transfer (AT)

Capacity gap impact on intermediate feature alignment



Forced spatial alignment acts as noise due to Student's depthwise convolution capacity limit.



Removing spatial constraints ( $\beta_s=0$ ) attenuates regression, as temporal dynamics are more transferable.

# Born-Again Networks & Knowledge Collapse

Iterative self-distillation MobileNet3D → MobileNet3D

$\alpha = 0.5, T = 2.5$

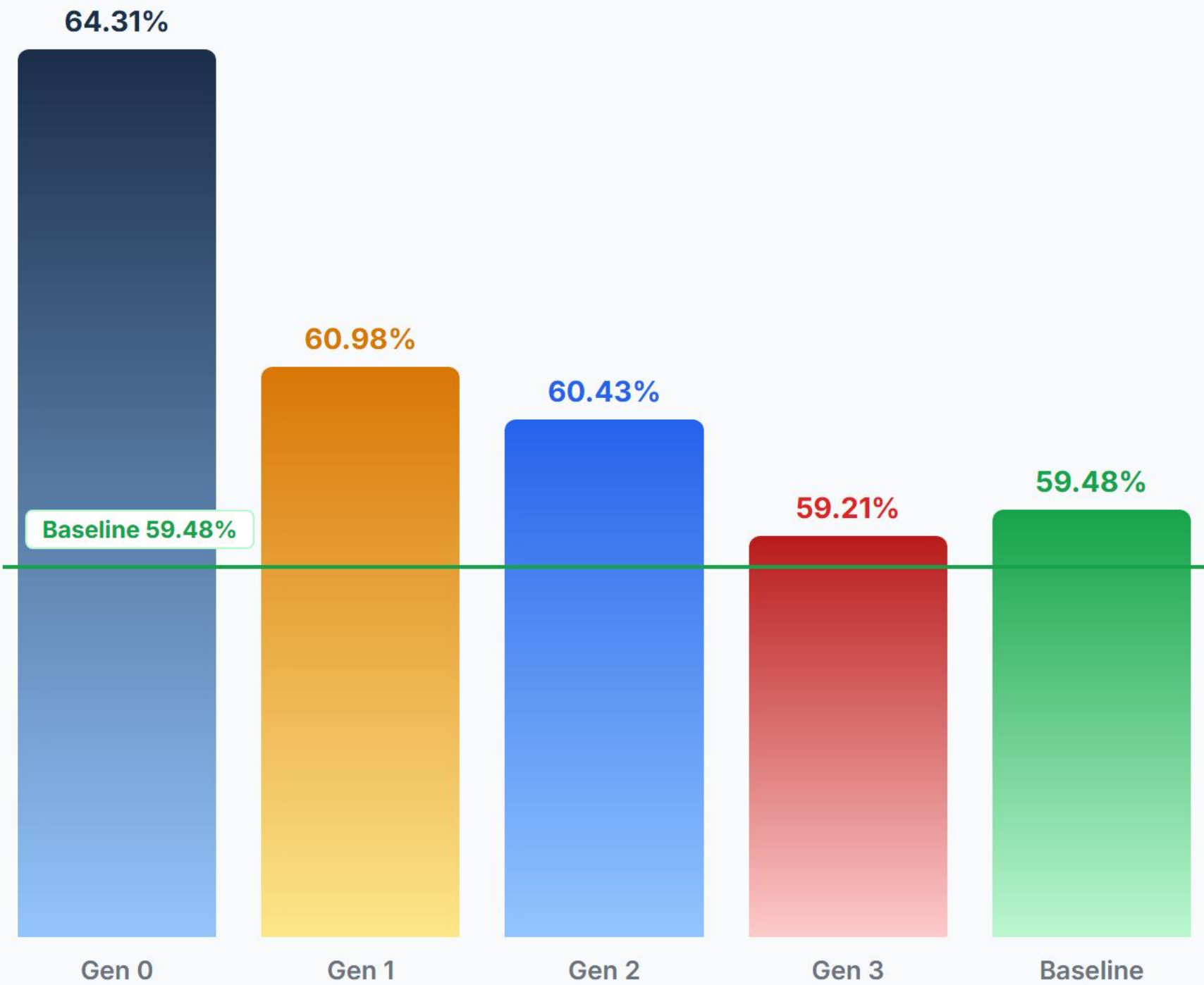
Iterative Self-Distillation

| GEN      | TEACHER              | T   | A   | ACCURACY |
|----------|----------------------|-----|-----|----------|
| Gen 0    | 3D ResNet-50         | 8.0 | 0.7 | 64.31%   |
| Gen 1    | MobileNet3D Gen 0    | 2.5 | 0.5 | 60.98%   |
| Gen 2    | MobileNet3D Gen 1    | 2.5 | 0.5 | 60.43%   |
| Gen 3    | MobileNet3D Gen 2    | 2.5 | 0.5 | 59.21%   |
| Baseline | No Teacher (Scratch) | —   | —   | 59.48%   |

Considerations

A compressed student (64.31%) lacks the visual representational capacity to act as an effective teacher. Distilling at **T=2.5** fails to prevent error propagation, causing progressive degradation back to baseline (59.48%).

ACCURACY ACROSS GENERATIONS

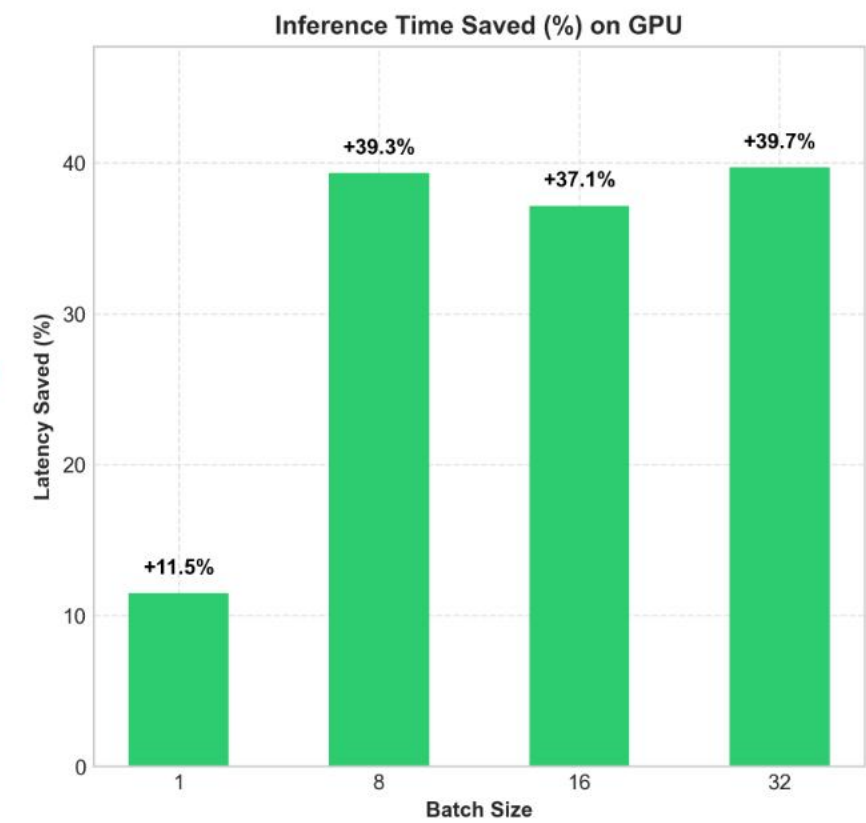
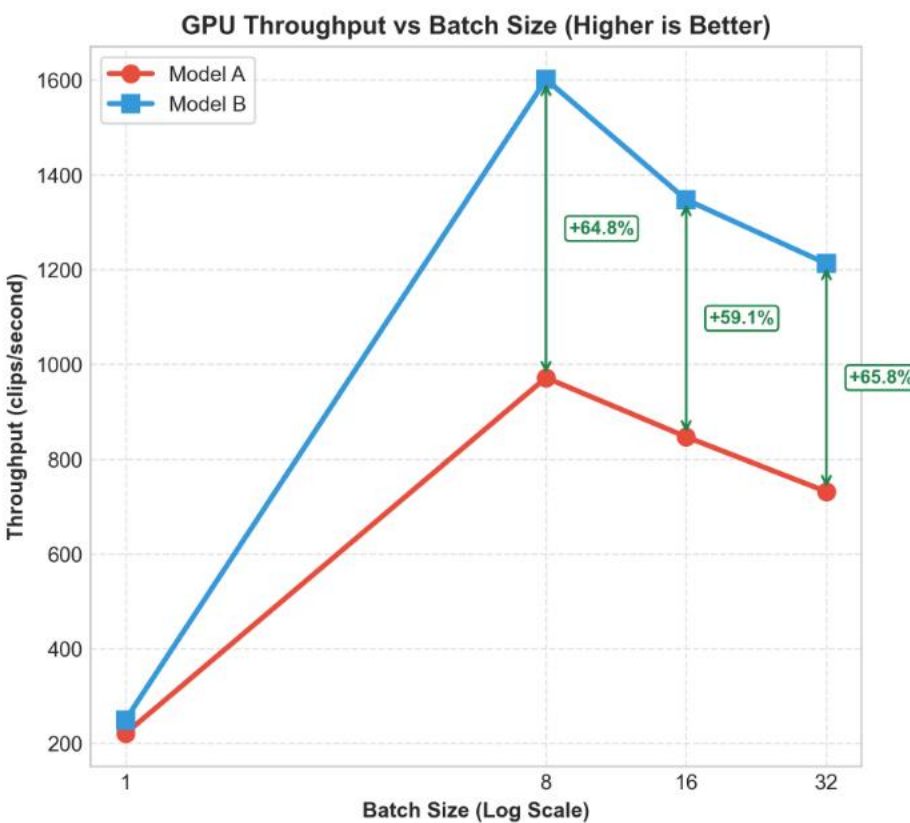
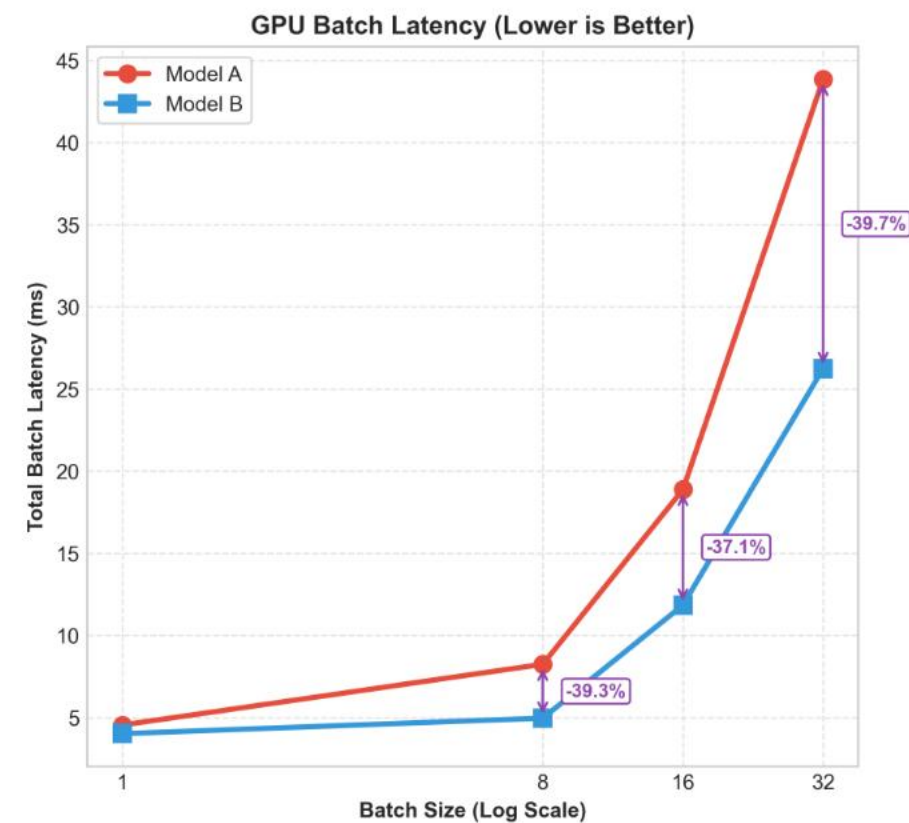




# Cross-Frame Distillation

Asymmetric temporal sub-sampling (T=20, a=0.7) — 16f Student vs 24f Teacher

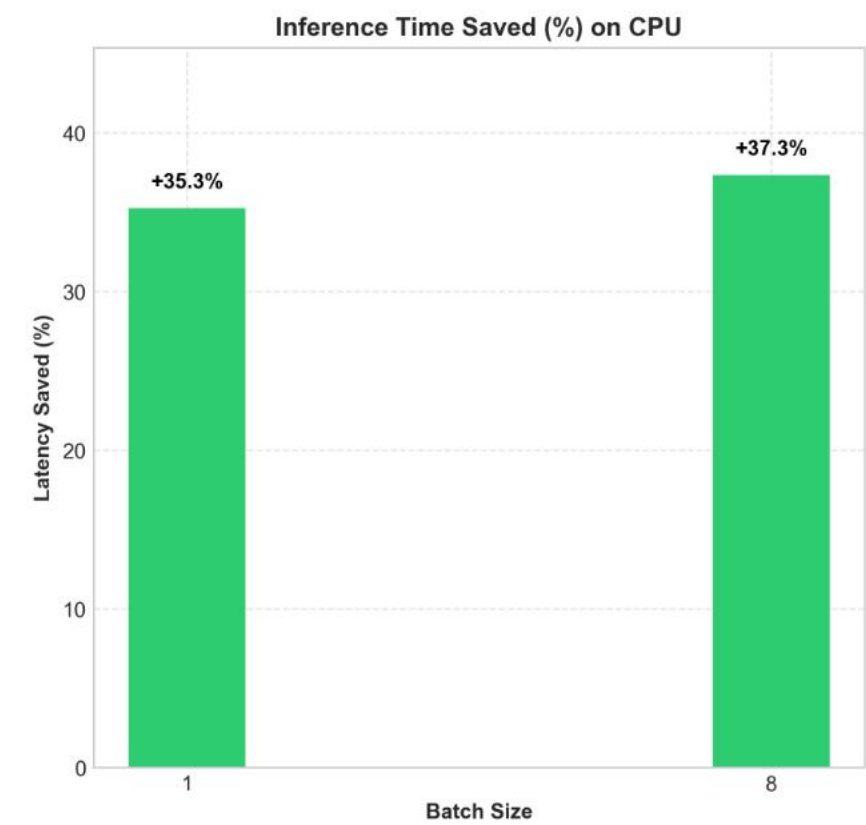
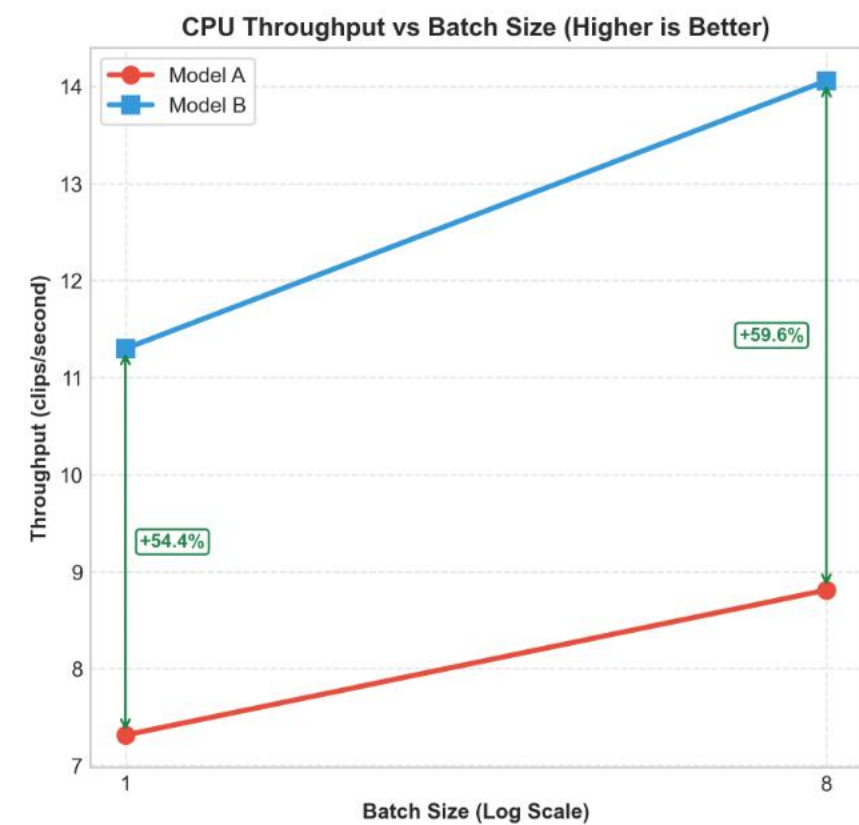
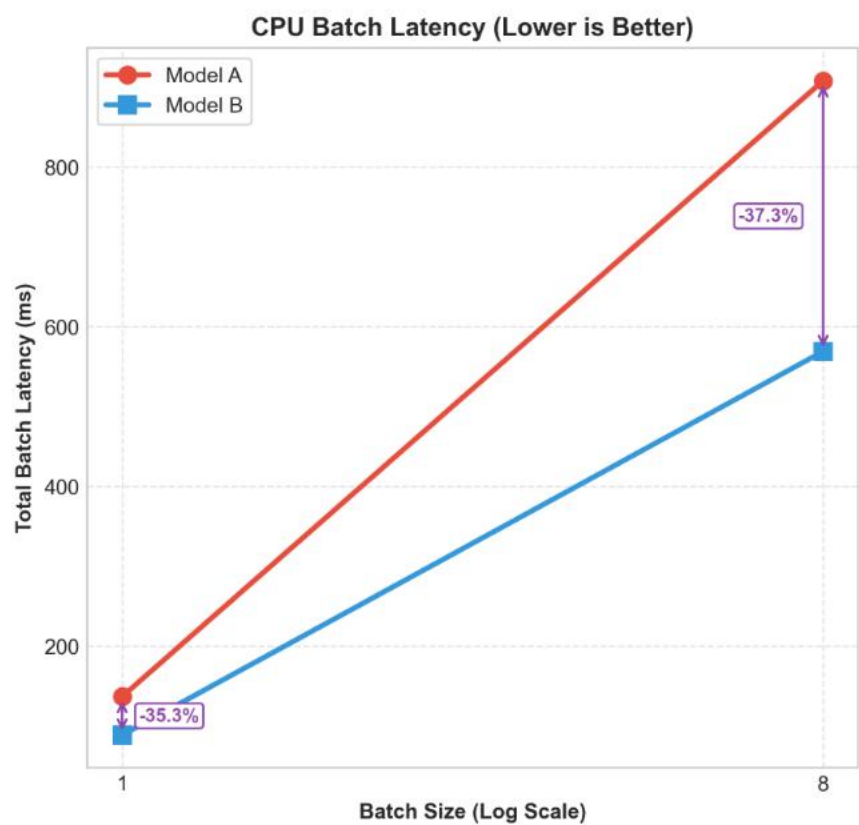
Test set inference Benchmark (GPU & CPU): Model A (24f) (Top-1: 65.85%) vs Cross Frame Model B (16f) (Top-1: 62.68%)



PERFORMANCE GAP  
**-3.17%**

65.85% (24f) → 62.68% (16f)

Minimal accuracy cost for temporal compression



INFERENCE SPEEDUP  
**2.11x**

CPU latency halved (-52.6%)

GPU throughput boosted by +68.8%



# Key Takeaways & Future Work

## NO FRAME DIST. (24F)

ACCURACY  
**65.85%** +6.37 pp  
vs. 59.48% Baseline

CPU LATENCY  
**53.52 ms**  
24-frame computational load

GPU THROUGHPUT  
**991 v/s**  
Standard throughput

## CROSS-FRAME DIST. (16F)

ACCURACY  
**62.68%** +3.20 pp  
vs. 59.48% Baseline

CPU LATENCY  
**25.36 ms** -52.6% (2.11×)  
Latency halved on Edge CPU

GPU THROUGHPUT  
**1673 v/s** +68.8%  
Register & Tensor Core alignment

### Lessons Learned

The spatial capacity gap limits intermediate feature transfer. Logit distillation at high temperature (T=20) is the most stable guide.

### Key Limitations

- (KD) High temperature (T=20) over-smooths fast, cyclic actions.
- (AT) Spatial capacity gap limits intermediate Feature Distillation.
- (BAN) Error propagation & bias accumulation in Born-Again Networks.

### Future Work

- **Multi-Scale Temporal Sampling:** Alternate global segmented sampling with local, tight-stride windows to capture both long-term history and fine-grained cyclic actions.
- **Flexible Feature Alignment:** Introduce temporary learnable adapter blocks (e.g. 1×1×1 convs) during training to bridge the AT capacity gap.



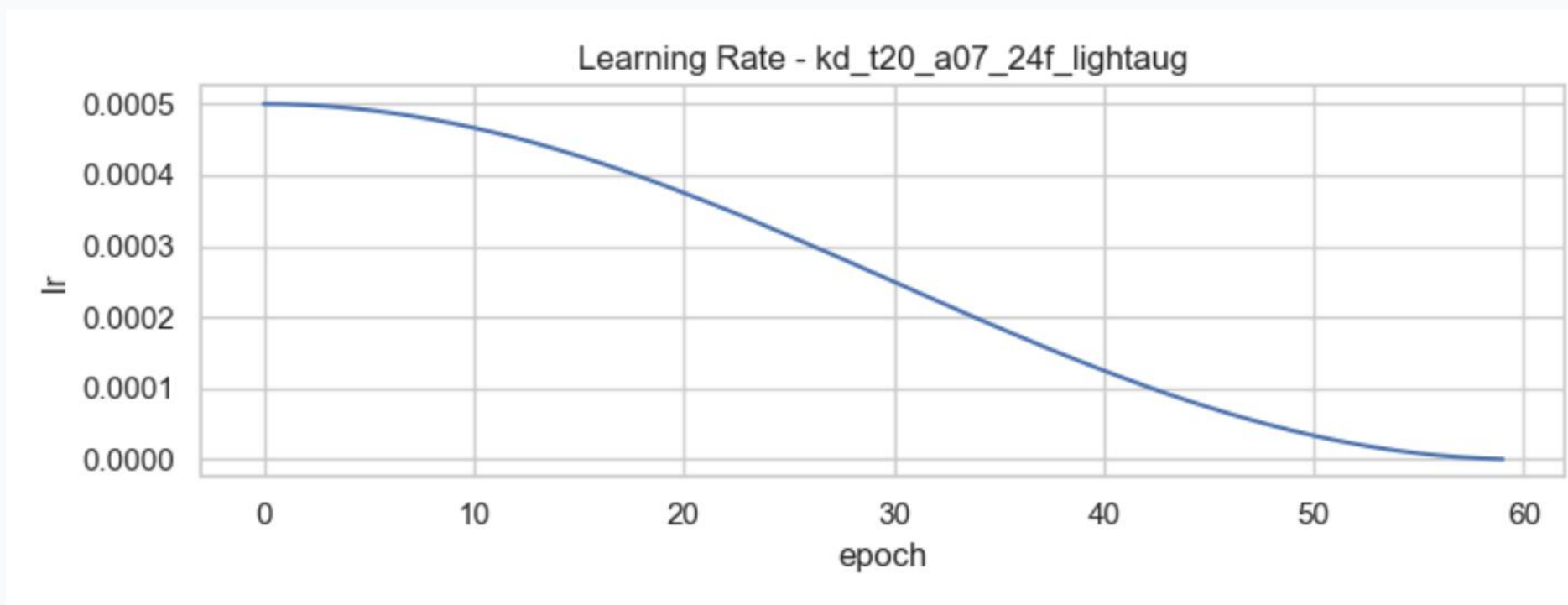
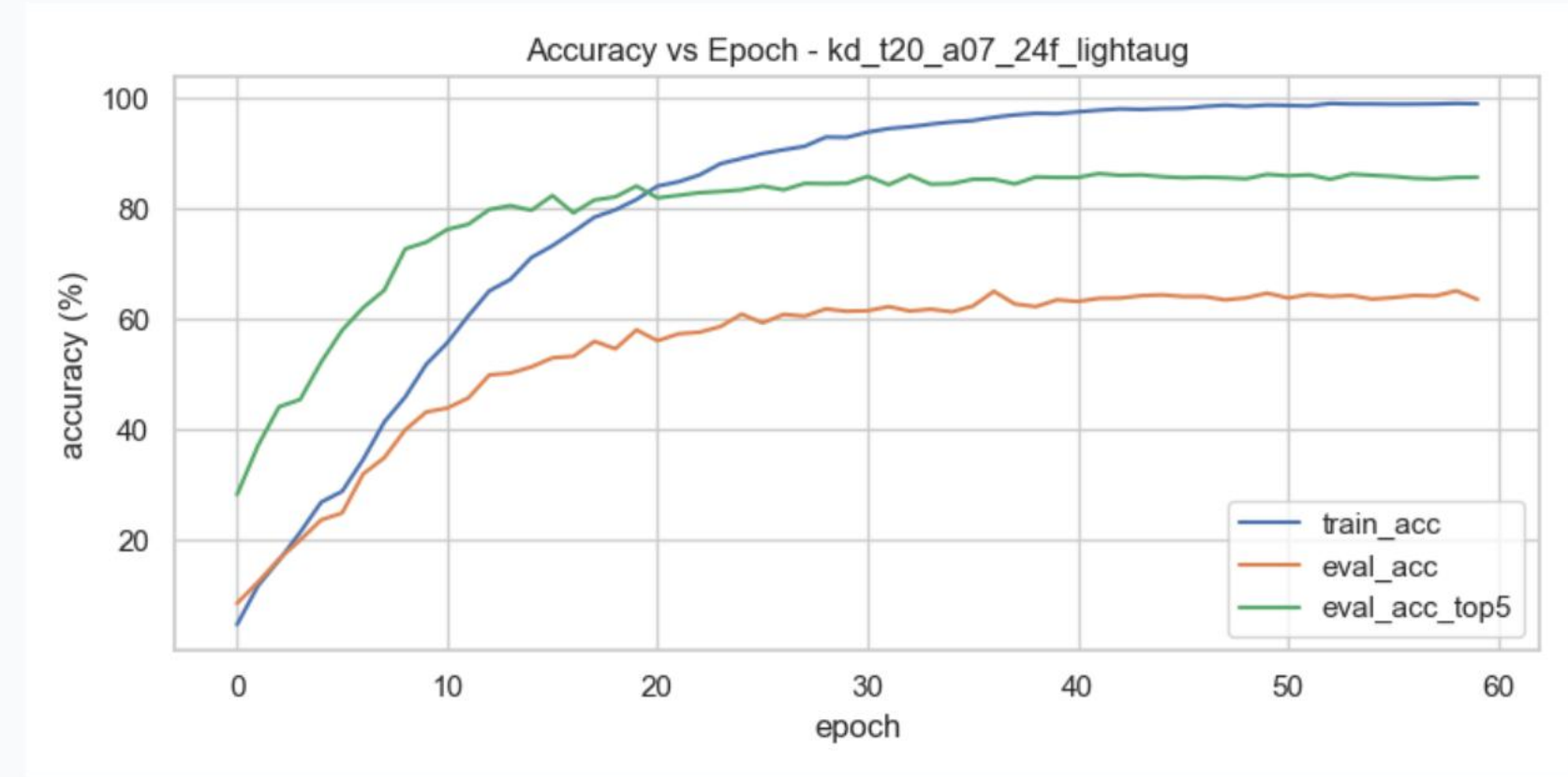
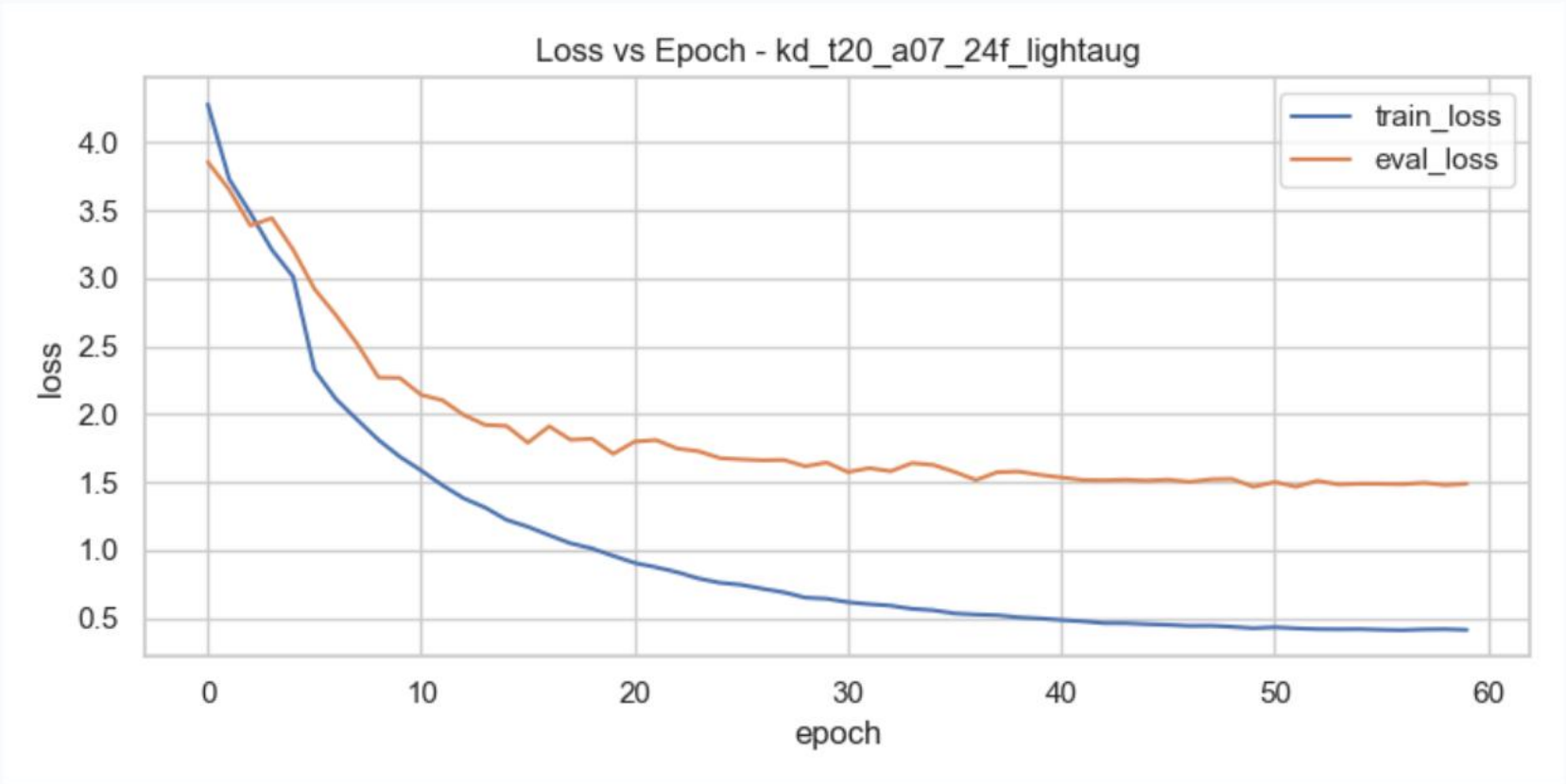
ADDITIONAL MATERIAL

# Backup Slides

Quantitative plots, qualitative detailed analyses, and CM predictions.

# Training Curves — KD T=20, a=0.7

Loss, accuracy, and LR schedule over 60 epochs



## Stability & Convergence

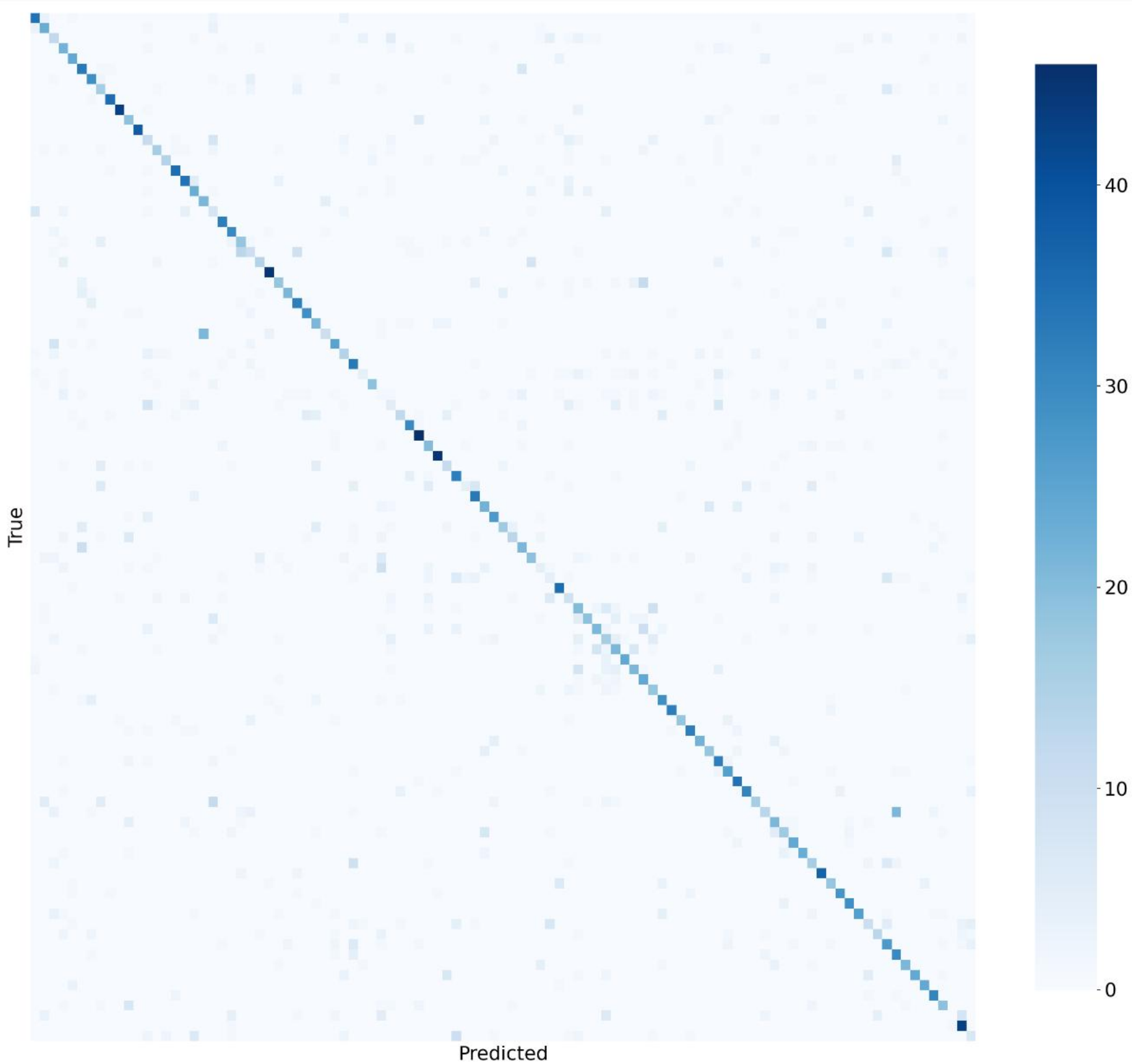
Stable training over 60 epochs. First 10 epochs serve as warmup with hard targets before soft distillation loss activates.



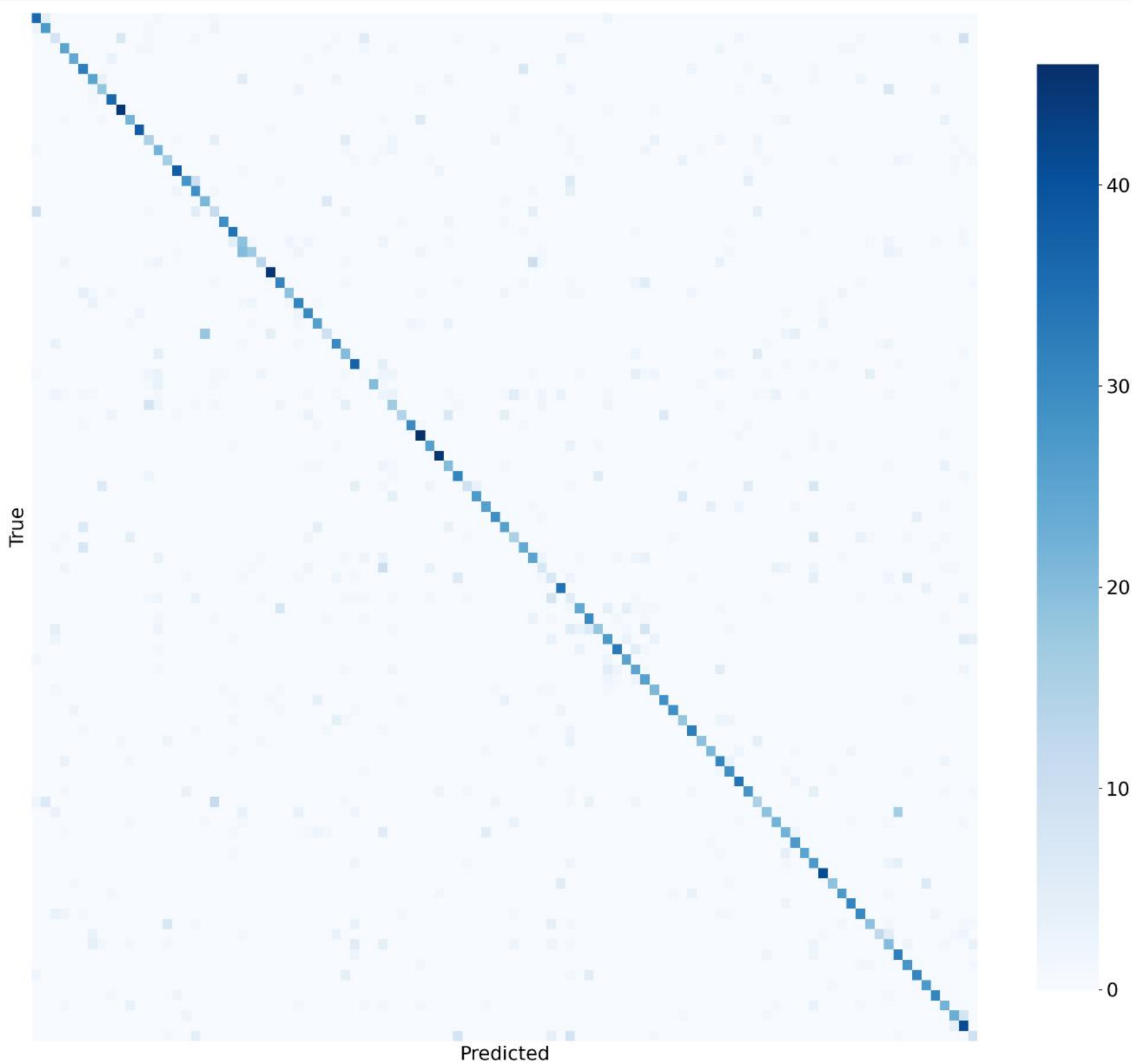
# Confusion Matrix & Error Analysis

Baseline vs KD T=20 - Full normalized matrices

BASELINE - 59.48%



KD T=20 - 65.85%



# Attention Transfer Loss: Before and After

Decomposing 3D video features to resolve the spatial-temporal collapse

## Before: Single Collapsed Loss

$$\mathcal{L}_{AT} = \sum_l || \text{normalize}(\text{mean}_c(F^2))_{flat} - ... ||^2$$

- **Block Mismatch:** Hooked ResNet stages [3, 4, 5]. Stage 5 is the classifier head (produces 2D logits [B, 101]), which caused runtime crashes and zero gradient contribution.
- **1D Interpolation Mismatch:** Resizing a flattened 1D vector of shape [T × H × W] treats space and time as a single coordinate line. This mathematically mixed pixels across frame boundaries and unrelated spatial points, destroying feature geometry.
- **Temporal Signal Dilution:** Normalizing the entire flattened space meant temporal peaks were washed out by spatial dimensions.

59.14%

## After: Spatial & Temporal Decomposition

$$\mathcal{L}_{AT} = \beta_s \mathcal{L}_{spatial} + \beta_t \mathcal{L}_{temporal}$$

- **Correct Block Pairing:** Hooked ResNet stages [2, 3, 4] (intermediate 3D features: [B, C, T, H, W]), correctly matching student stages [2, 4, 6].
- **2D Bilinear Interpolation:** Spatial attention is calculated as mean<sub>c,t</sub>(F<sup>2</sup>) and resized spatially via 2D bilinear interpolation, preserving frame layout.
- **Temporal AT Isolation:** Temporal attention is calculated as mean<sub>c,h,w</sub>(F<sup>2</sup>) yielding [B, T], which is transferred directly without any interpolation noise.

64.58% · 65.03%

Symmetric: β<sub>s</sub>=0.05, β<sub>t</sub>=0.05 · Temporal: β<sub>s</sub>=0, β<sub>t</sub>=0.10



# Analysis of Attention Transfer Limitations

Theoretical barriers of classical attention transfer in lightweight student training

## 1. Attention Loss & Information Collapse

### Key Bottleneck: Channel Pooling

To align mismatched feature map dimensions, classical AT averages activation maps along the channel axis. This mathematical flattening results in a severe loss of representation:

- **Feature Flattening:** Collapsing the channel dimension merges specialized filters (textures, edges, motion directions) into a generic spatial-temporal map.
- **Loss of Semantics:** The student is forced to learn *where* or *when* to look, but misses the underlying semantic explanation (the *why*).
- **Coarse Regularization:** The student mimics a highly-abstracted projection rather than the rich visual filters that drive the teacher's decisions.

## 2. Capacity Gap & Structural Mismatch

### Key Conflict: Geometric Incompatibility

The student model's architectural constraints limit its capacity to mimic the teacher's representations due to fundamental differences in conv designs:

- **Convolutional Disparity:** The teacher utilizes standard 3D convolutions that extract joint space-time-channel correlations.
- **Depthwise Factorization:** The student (MobileNet3D) uses Depthwise Separable Convolutions, which strictly isolate spatial-temporal filters from channel mixtures.
- **Gradient Conflict:** Forcing the student to replicate complex attention maps creates an over-constrained optimization. This leads to conflicting gradients (noise) that degrade overall convergence.